

House Price Prediction using TensorFlow

This project uses machine learning and deep learning (TensorFlow/Keras) to predict house prices using numeric and categorical features.

The model is built using a regression neural network with data preprocessing, scaling, and evaluation.

Project Overview

This project demonstrates:

- Data loading & preprocessing
- Handling categorical and numeric data
- Feature engineering
- Train/validation split
- Scaling using StandardScaler
- Building a TensorFlow regression model
- Model training, validation & testing
- Saving and loading model
- Making predictions

Dataset Structure

The dataset contains columns such as:

- `sqft` — Size of the house
- `bedrooms`
- `bathrooms`
- `location`
- `price` (target)

Place your dataset in:

```
'''
```

```
/content/train.csv
```

```
/content/test.csv
```

```
'''
```

Installation

Install required packages:

```
'''
```

```
pip install tensorflow pandas scikit-learn numpy matplotlib
```

```
'''
```

Data Preprocessing

Steps performed:

- Removing missing values
- Encoding categorical columns using `OneHotEncoder`
- Splitting dataset (train/validation)
- Scaling numeric features using `StandardScaler`

TensorFlow Model Architecture

A simple regression neural network:

```
'''
```

```
Dense(128, activation='relu')
```

```
Dense(64, activation='relu')
```

```
Dense(32, activation='relu')
```

```
Dense(1) # price output
```

Training

Compile the model:

```
'''  
  
model.compile(optimizer='adam', loss='mse', metrics=['mae'])  
  
'''
```

Train:

```
'''  
  
model.fit(x_train, y_train, epochs=50, validation_data=(x_val, y_val))  
  
'''
```

Saving Model

Use:

```
'''  
  
model.save("house_price_model.keras")  
  
'''
```

Evaluation

Visualize and evaluate:

- Mean Absolute Error (MAE)
- Loss curves

Prediction

```
'''  
  
loaded_model = tf.keras.models.load_model("house_price_model.keras")  
  
pred = loaded_model.predict(test_scaled)  
  
'''
```

GitHub Instructions

1. Create a new repository
2. Upload:
 - All datasets
 - Notebook / Python script
 - Saved model
 - README file
3. Commit & push

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You can modify this file as needed.